DS607 GROUP PROJECT PROPOSAL

Can a Simple Operational Benchmark Outperform SARIMA in Forecasting China's Headline and Core Inflation?

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INFERENCE QUESTION

Does a fitted seasonal time-series model produce more accurate forecasts of mainland China's headline and core consumer price inflation than a simple operational benchmark based on recent calendar-month inflation averages? The project will also examine whether explicitly accounting for the moving timing of Chinese New Year improves headline-inflation forecasts.

DATA SOURCE AND SERIES

Monthly mainland China Consumer Price Index data will be obtained from CEIC Data through the SMU Libraries subscription. The common headline and core sample begins in January 2006 and extends through the latest available observation. The principal modelling series are published month-on-month headline and core inflation. Forecast monthly price-index paths will be compounded and converted into year-on-year inflation forecasts for evaluation against the corresponding published year-on-year series. Public Gregorian-lunar calendar data will be used to construct known Chinese New Year moving-holiday regressors.

METHODS

The operational benchmark recursively forecasts monthly inflation using recent historical averages for each calendar month. Statistical challengers will use parsimonious SARIMA models selected within a pre-2015 development sample. A headline SARIMAX extension will include fixed calendar-month effects and separate moving-holiday regressors for Chinese New Year demand pressure and post-holiday normalisation. Model orders and evaluation rules will be fixed before final evaluation. Expanding-window forecasts will be generated monthly at horizons up to 24 months and compared using paired mean error, mean absolute error, and root mean squared error. Convergence and the ability to produce complete, valid forecast trajectories will also be reported.

EXPECTED OUTPUTS

The project will produce horizon-specific comparisons of forecast accuracy for headline and core inflation, diagnostic evidence on model stability and forecast reliability, and an assessment of whether statistical complexity and China-specific calendar knowledge materially improve upon the operational benchmark.

LICENSING NOTE CEIC data are accessed through the restricted SMU Libraries subscription and will not be redistributed. The submitted work will report derived statistics, model outputs, and forecast-evaluation results rather than reproduce the licensed source dataset.